3.6 Sensitivity analysis in Linear Programming

Evaluate the "sensitivity" of an optimal solution with respect to variations in the model parameters.

Example: Production planning

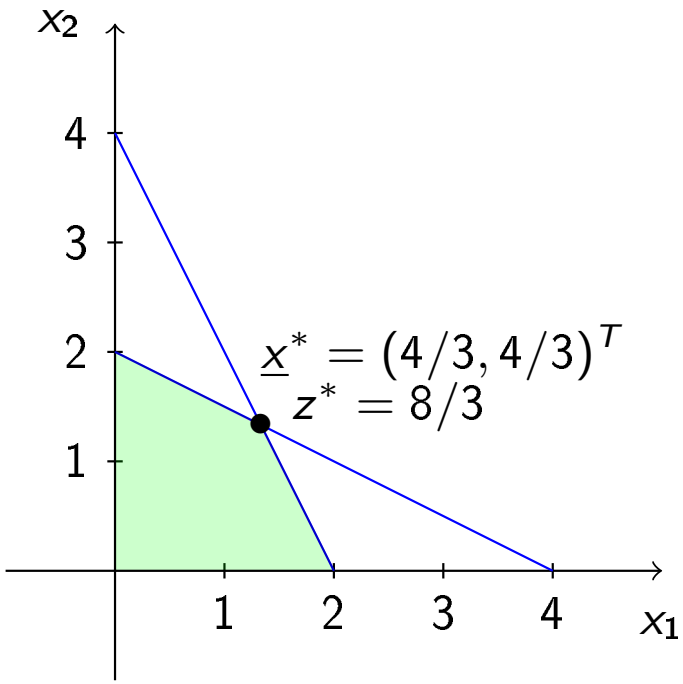
$$
\begin{array}{lll}
\max & \displaystyle\sum_{j=1}^{n} p_j x_j & \\
\text{s.t.} & \displaystyle\sum_{j=1}^{n} a_{ij} x_j \leq b_i, & i = 1, \ldots, m \\
& x_j \geq 0, & j = 1, \ldots, n
\end{array}
$$

- p_j is the unit profit of j-th product
- b_i is the availability of i-th resource

3.6.1 Geometric interpretation

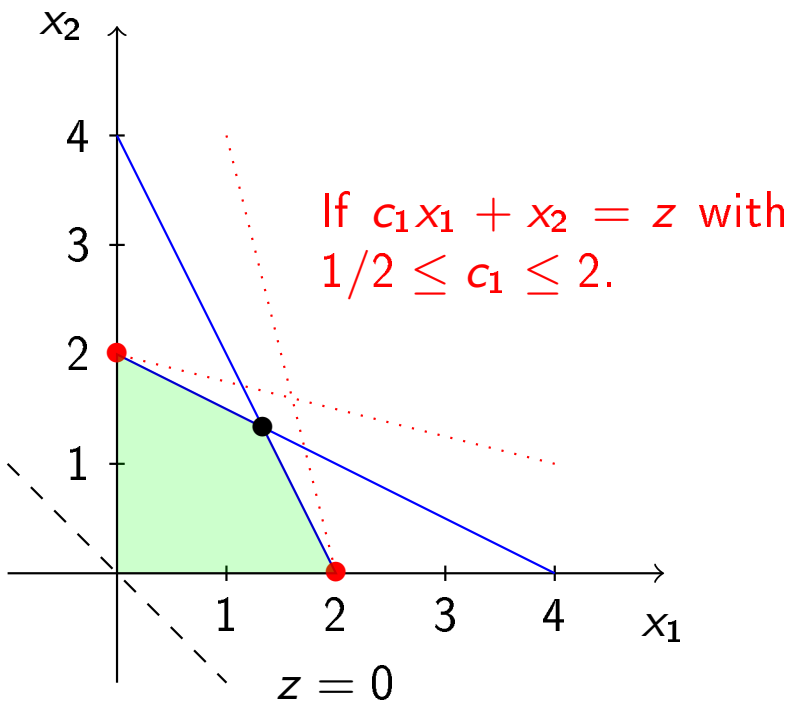
Example:

$$
\begin{array}{ll}
\max & x_1 + x_2 \\
\text{s.t.} & x_1/2 + x_2 \leq 2 \\
& 2x_1 + x_2 \leq 4 \\
& x_1, x_2 \geq 0
\end{array}
$$



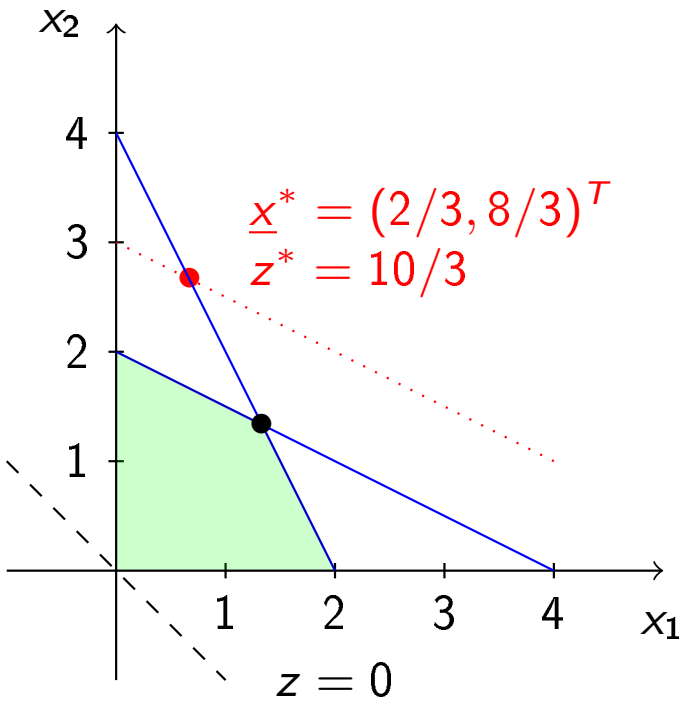
Variation of the objective function coefficients

$$\begin{array}{ll} \max & c_1 x_1 + x_2 \\ \text{s.t.} & x_1/2 + x_2 \leq 2 \\ & 2x_1 + x_2 \leq 4 \\ & x_1, x_2 \geq 0 \end{array}$$



Variations of the right-hand side terms

$$\begin{array}{ll} \max & x_1 + x_2 \\ \text{s.t.} & x_1/2 + x_2 \leq 2 + 1 \\ & 2x_1 + x_2 \leq 4 \\ & x_1, x_2 \geq 0 \end{array}$$



Definition 23

The shadow price of the i-th resourse is the maximum price the company is willing to pay to buy an additional unit of i-th resource.

Shadow price of the first resource $= 10/3 - 8/3 = 2/3$

3.6.2 Sensitivity analysis: algebraic form

Evaluate the "sensitivity" of an optimal solution when the model parameters (c_j, a_{ij}, b_i) vary.

Given

$$(P) \quad \begin{array}{ll} \min & \underline{c}^T \underline{x} \\ \text{s.t.} & A\underline{x} = \underline{b} \\ & \underline{x} \geq \underline{0} \end{array}$$

and optimal basic feasible solution $\underline{x}^*$:

$$\begin{array}{l} \underline{x}_B^* = B^{-1}\underline{b} \geq \underline{0} \\ \underline{x}_N^* = \underline{0} \end{array}$$

Within which limits the basis B remains optimal?

Conditions:

1. $B^{-1}\underline{b} \geq \underline{0}$ feasibility
2. $\underline{\bar{c}}_N^T = \underline{c}_N^T - \underline{c}_B^T B^{-1} N \geq \underline{0}^T$ optimality

Study the next slides before the **computer lab assignment**.

Variation of right-hand-side terms

$\underline{b}' := \underline{b} + \delta_k \underline{e}_k$ with $\underline{e}_k$ the unitary vector equal to 1 on the k-th position, $1 \leq k \leq n$.

The basis B with the basic feasible solution

$$\underline{x}^* = \begin{bmatrix} B^{-1}(\underline{b} + \delta_k \underline{e}_k) \\ \underline{0} \end{bmatrix} \qquad \underline{\bar{c}}_N \text{ does not vary}$$

remains optimal as long as

$$B^{-1}(\underline{b} + \delta_k \underline{e}_k) \geq \underline{0} \Rightarrow B^{-1}\underline{b} \geq -\delta_k B^{-1}\underline{e}_k$$

These m inequalities define an interval of variation for δ_k.

B remains optimal, but the optimal basic feasible solution changes.

The objective function values goes from $\underline{c}_B^T B^{-1}\underline{b}$ to $\underline{c}_B^T B^{-1}(\underline{b} + \delta_k \underline{e}_k)$, thus

$$\Delta z^* = \underbrace{\underline{c}_B^T B^{-1}}_{\underline{y}^{*T}}(\delta_k \underline{e}_k) = \underbrace{\delta_k \overbrace{y_k^*}^{\frac{\partial z^*}{\partial b_k}}}_{\text{shadow price}} \quad (y_k^* \text{ optimal value of the dual variable})$$

Variation of the cost coefficients

Given $\underline{c}' := \underline{c} + \delta_k \underline{e}_k$ a basis B remains optimal as long as

$$\overline{\underline{c}}_N'^T = \underline{c}_N'^T - \underline{c}_B'^T B^{-1} N \geq \underline{0}.$$

In such a case, the optimal basic (feasible) solution does not change:

$$\underline{x}_B^* = B^{-1} \underline{b} \quad \text{and} \quad \underline{x}_N^* = \underline{0}.$$

Variation of the cost coefficients

- If x_k is a <u>nonbasic variable</u> $\quad \underline{\bar{c}}_N'^T = \underline{c}_N'^T - \underline{c}_B'^T B^{-1} N \geq \underline{0}^T$ with $\underline{c}_B'^T = \underline{c}_B^T$

$$\begin{aligned} \underline{\bar{c}}_N'^T &= (\underline{c}_N^T + \delta_k \underline{e}_k^T) - \underline{c}_B^T B^{-1} N \\ &= (\underline{c}_N^T - \underline{c}_B^T B^{-1} N) + \delta_k \underline{e}_k^T \\ &= \underline{\bar{c}}_N^T + \delta_k \underline{e}_k^T \geq \underline{0}^T \\ &\Rightarrow \delta_k \geq -\bar{c}_k \end{aligned}$$

Reduced cost = max decrease of c_k for which B remains optimal (larger decrease $\Rightarrow \bar{c}_k' < 0$)

$$\Delta z^* = 0 \text{ since } z^* = \underline{c}_B^T B^{-1} \underline{b} = \underline{c}_B^T \underline{x}_B^*$$

- If x_k is a <u>basic variable</u> $\quad \underline{\bar{c}}_N'^T = \underline{c}_N'^T - \underline{c}_B'^T B^{-1} N \geq \underline{0}^T$ with $\underline{c}_N'^T = \underline{c}_N^T$

$$\begin{aligned} \underline{\bar{c}}_N'^T &= \underline{c}_N^T - (\underline{c}_B^T + \delta_k \underline{e}_k^T) B^{-1} N \\ &= (\underline{c}_N^T - \underline{c}_B^T B^{-1} N) - \delta_k \underline{e}_k^T B^{-1} N \\ &\Rightarrow \underline{\bar{c}}_N^T - \delta_k \rho_k^T \geq \underline{0}^T \text{ (where } \rho_k^T \text{ is the } k\text{-row of } B^{-1}N) \\ &\Rightarrow \underline{\bar{c}}_N^T \geq \delta_k \rho_k^T \quad (n-m \text{ inequalities define a variation interval for } \delta_k) \end{aligned}$$

$$\Delta z^* = \delta_k \underline{e}_k^T B^{-1} \underline{b} = \delta_k \underbrace{x_k^*}_{\frac{\partial z^*}{\partial c_k}}$$

Example

Initial tableau:

$$\begin{array}{ll} \min & \underline{c}^T \underline{x} \\ \text{s.t.} & A\underline{x} = \underline{b} \\ & \underline{x} \geq \underline{0} \end{array}$$

		x_1	x_2	x_3	x_4	x_5	x_6
$-z$	0	-3	-1	-3	0	0	0
x_4	2	2	1	1	1	0	0
x_5	5	1	2	3	0	1	0
x_6	6	2	2	1	0	0	1

Optimal tableau:

		x_1	x_2	x_3	x_4	x_5	x_6
$-z$	$27/5$	0	$7/5$	0	$6/5$	$3/5$	0
x_1	$1/5$	1	$1/5$	0	$3/5$	$-1/5$	0
x_3	$8/5$	0	$3/5$	1	$-1/5$	$2/5$	0
x_6	4	0	1	0	-1	0	1

Determine the dual optimal solution, the variations of $\underline{b}$ and the variations of $\underline{c}$.